Tascam 38 amplifier card - bench worksheet

Checks that the schematics, datasheets and manual cannot settle. Fill in each reading, tick whether it matches the expected value, and report the readings back so they can be recorded in `docs/tascam-38/circuit-analysis.md`. IDs (A1, C17, ...) refer to that document.

Date _____ Machine serial _____ Meter used _____

Card measured (channel / slot) _____ Meter leads shorted reads (ohms) _____ Initials _____

Before you start

Continuity checks are with the machine **unpowered** (mains disconnected). Card checks are on a card removed from the motherboard. The bias / erase circuit (J108-2, the U103 output) carries tens of volts AC at 150 kHz while recording - do not probe it with the record circuit live. Confirm pin 1 on each physical connector from the board silkscreen before measuring: the schematic numbers J105 / P105 from the opposite end to the other card connectors. Subtract the shorted-lead reading from low-ohm results.

1. On a removed card, unpowered - ohmmeter; finishes the relay question (A1)

#	Measure between	Expect	Reading	Match?	Notes	ID
-	J101-4 and J101-3	low (TR101 primary)	0 ohm (done)	yes	recorded 2026-10-13	A1
-	J106-7 and TP1	open	open (done)	yes	recorded 2026-10-13	A1
1a	J108-1 and J108-3	open		☐		A1
1b	J108-1 and TP1	near 0 ohm		☐		A1

If 1a reads near 0 and 1b open, K103's lower set is drawn released, not energised - note it and stop to report.

2. Motherboard, power off, all cards out - ohmmeter

#	Measure between	Expect	Reading	Match?	Notes	ID
2a	P108-1 and P171-7 (0V)	0 ohm		☐		C17 A10
2b	P182-1 and P182-2 (MASTER OSC connector)	open		☐		M4
2c	P104-5 and P171-11 (0V)	open		☐		C18
2d	P171-9 (+B 15V) and channel-2 P102-2 (ch2 line out)	open		☐		C19

2e. Contacts with no wire drawn - motherboard, power off; C08, C09

Channel-1 motherboard sockets. Measure each against every rail on P171; expect **open** everywhere. Write the reading, or tick if open.

Contact	0V (P171-7)	24V (P171-8)	+B (P171-9)	-B (P171-10)	6V (P171-1)	Any other net found
P102-1						
P102-6						
P102-7						
P103-1						
P103-3						
P104-7						
P105-2						
P105-3						
P105-4						
P107-5						
P107-6						

On a removed card, J102-1 and J102-6 should each read near 0 ohm to J101-3 (the card common bus):

Measure between	Expect	Reading	Match?
J102-1 and J101-3	near 0 ohm		☐
J102-6 and J101-3	near 0 ohm		☐

3. Head connectors - inspection plus ohmmeter; C12, C20

The schematic gives no pin numbers for P121 / P151 / P161, only "upper" and "lower" contacts. For each, record the physical pin number, wire colour, and what it reaches (winding end, shield). Expected card mapping in brackets.

Connector	Contact on drawing	Physical pin no.	Wire colour	Goes to (winding / shield)	Winding resistance
P121 repro	upper (J101-4)				
P121 repro	lower (J101-3)				
P151 sync	lower (J106-7)				
P151 sync	upper (J106-8)				
P161 erase	upper (J108-2)				
P161 erase	lower (J108-3)				
Shield	head-board shield bus (J401-J403)				

4. Inspection - no meter

#	Check	Result	ID
4a	OUT SELECT buttons: do they latch? ☐ latch ☐ momentary		M5
4b	Can two OUT SELECT buttons stay pressed together? ☐ yes ☐ no (interlocked)		M5
4c	R139 colour code (card, near Q108)	68 ohm (done, photo)	A7
4d	L108 marking / value (card, erase feed)		A5
4e	Jumpers fitted: ☐ W109 ☐ W110 ☐ W112 ☐ W119 ☐ W132		A8
4f	U103 module marking (bias amp; manual: TEAC 5292210600)		A4
4g	Any visible component differences between channel cards		C10

5. Optional - machine powered - DC voltmeter to 0V; confirms the datasheet reading

Transport state	P171-6 PLAY MUTE expect	Reading	P171-5 REC SIGNAL expect	Reading	Match?
STOP	about +5.3 V		open / ~0 V		☐
PLAY	about -6.8 V		open / ~0 V		☐
FF or REW	about +5.3 V		open / ~0 V		☐
RECORD, one track armed	about -6.8 V		about 24 V		☐
PAUSE (from PLAY)	about +5.3 V		open / ~0 V		☐
Record-pause	about +5.3 V		about 24 V		☐

#	Measure	Expect	Reading	ID
5a	P171-3 POWER MUTE, settled after power-on (and roughly how long it takes)	near 0 V, ~1.8 s		M6
5b	Q118 base on a card, in RECORD (armed) and in PLAY	unknown		A3
5c	Power supply terminal 41 ("+2V"), counter connected	unknown		M9
5d	Scope, if available: bias frequency at J182-8	150 kHz		A5

Notes